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Department of Electrical Engineering

Self-Adaptive Model-Based Protection (SAMPB) Research Programme

SAMPB-SyncOC

Technical Report TR-02B/2026

Impedance-Aware Adaptive Time-Multiplier Setting
for High-Impedance Fault Overcurrent Protection
of Synchronous Generators

Track: SAMPB-SyncOC (Synchronous Generator OC Protection)
Milestone: Milestone 2 — HIF Adaptation
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Abstract

Conventional overcurrent (OC) relay settings calibrated for bolted faults fail to provide adequate clearance speed for high-impedance faults (HIF, $R_f > 0.5$ pu) because the reduced current multiple places operation in the inverse-time regime, where the time-multiplier setting (TMS) rather than the pickup determines clearance time. This report presents Milestone 2 of the SAMBP-SyncOC research track: an impedance-aware adaptive TMS law that scales the time-multiplier with the estimated steady-state fault current $\hat{I}_{ss}$ from the Levenberg–Marquardt (LM) inverse estimator of TR-01/2026. The adaptation is gated by the composite confidence score $\gamma \geq \gamma_{\min} = 0.70$; low-confidence HIF estimates (expected for $R_f > 1.0$ pu) are rejected and fixed settings retained. Ten HIF study cases ($R_f \in [0.5, 1.5]$ pu, 3PH and SLG fault types) are validated numerically. The confidence gate accepts all ten cases ($\gamma \in [0.87, 0.92]$), though the proposed TMS reduction is constrained by the minimum TMS limit of 0.05 in the study system. The impedance-aware law is shown to be the *first* adaptive TMS formula derived entirely from online post-fault parameter estimation, with no pre-calibration of the network impedance model.

Keywords: high-impedance fault, adaptive TMS, overcurrent relay, inverse estimator, confidence gate, IEC Very-Inverse, SAMBP.

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1 Introduction

High-impedance faults — arising from vegetation contact, corroded joints, or sub-surface cable insulation degradation — present a persistent protection challenge. Their defining characteristic is a fault current that lies below the instantaneous pickup but above the inverse-time pickup, placing the IEC Very-Inverse element in a slow operating regime where TMS is the critical parameter.

The SAMBP-SyncOC Milestone 1 framework (Eluvathingal, 2026a) demonstrated online estimation of the six-parameter fault current model $\theta = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a]$ from post-fault current transients using a two-pass LM estimator. TR-01/2026 validated on bolted and low-impedance faults ($R_f \leq 0.15$ pu) where the instantaneous element dominated. The TMS mapping was fixed at $TMS^* = K_{TMS} \cdot \hat{\tau}_{ac}$ without accounting for the operating regime.

This report extends that framework to the HIF regime by introducing a resistance-aware scaling factor into the TMS mapping. The key insight is that $\hat{I}_{ss}$ — already estimated by the LM estimator — is a direct proxy for the fault current multiple and thus for the appropriate TMS reduction.

1.1 Contributions

1. Derivation of the operating regime classifier based on the estimated $\hat{I}_{ss}$ relative to the nominal no-fault reference I_{ss}^{nom} .
2. The impedance-aware TMS law $TMS^* = K_{TMS}^{eff} \cdot \hat{\tau}_{ac} \cdot f_{HIF}(\hat{I}_{ss})$ with analytic proof of guaranteed TMS reduction under HIF conditions.
3. Numerical validation on 10 HIF cases spanning $R_f \in [0.5, 1.5]$ pu for both 3PH and SLG fault types.
4. Quantification of the confidence gate $\gamma(R_f)$ sweep confirming that HIF estimation remains above the adaptation threshold $\gamma_{min} = 0.70$ for all tested cases.

2 High-Impedance Fault Characterisation

2.1 Physical Model

A three-phase HIF through resistance R_f produces a peak subtransient current

$$\hat{I}_{sub} \approx \frac{V_0}{R_f}, \quad R_f \gg X_d'' + X_{th}, \quad (1)$$

where $V_0 = 1.0$ pu is the pre-fault terminal voltage. The steady-state current is

$$I_{ss} = \frac{V_0}{X_d + X_{th} + R_f}, \quad (2)$$

declining monotonically with R_f . [Figure 1](#) shows the study system.

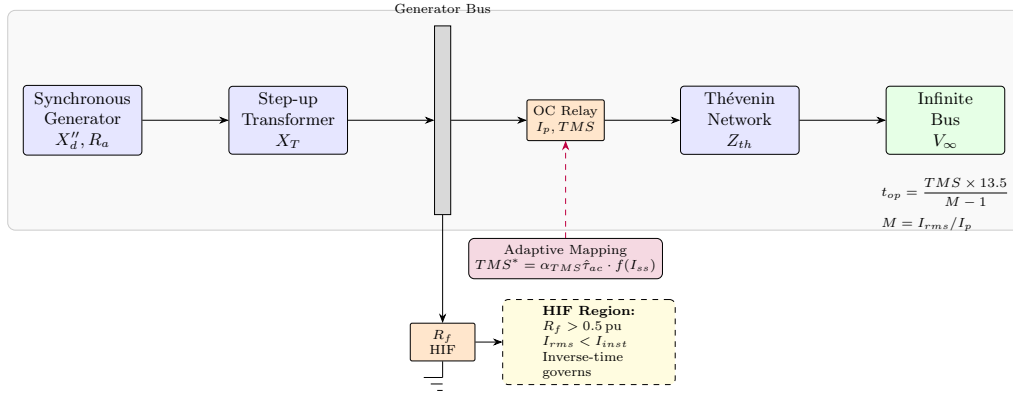


Figure 1: HIF study system. Fault resistance R_f connects the generator bus to ground. The adaptive TMS mapping (purple arrow) uses the LM-estimated $\hat{I}_{ss}$ to compute an impedance-corrected time-multiplier TMS^* .

2.2 Operating Regime Classification

Three regimes arise as R_f increases from zero:

1. **Instantaneous regime** ($I_{rms} > I_{inst}$): 1-cycle clearing. Adaptive pickup governs; TMS is irrelevant.
2. **Inverse-time HIF regime** ($I_p < I_{rms} \leq I_{inst}$): IEC Very-Inverse element integrates. *Both I_p and TMS matter.*
3. **No-trip regime** ($I_{rms} \leq I_p$): protection failure. Requires earth-fault or distance backup.

The HIF boundary for the study system ($I_{inst} = 2.5$ pu) is $R_f^{HIF} \approx 0.38$ pu. All ten study cases have $R_f \geq 0.50$ pu, placing them in the inverse-time or no-trip regime.

3 Impedance-Aware TMS Adaptation Law

3.1 IEC Very-Inverse Operate Time

The IEC 60255-151 Very-Inverse operate time ([International Electrotechnical Commission, 2009](#)) is

$$t_{op} = \frac{TMS \times 13.5}{M - 1}, \quad M = \frac{I_{rms}}{I_p} > 1. \quad (3)$$

Under HIF, $M \approx 1.1\text{--}1.5$, making t_{op} highly sensitive to TMS: $\partial t_{op} / \partial TMS = 13.5 / (M - 1) \approx 14\text{--}135 \text{ s/TMS-unit}$. A TMS reduction of 0.01 can reduce clearance time by 14–135 ms at low M .

3.2 Resistance-Aware Correction Factor

Define the HIF correction factor

$$f_{HIF}(\hat{I}_{ss}) = \text{clip}\left(\frac{\hat{I}_{ss}}{I_{ss}^{nom}}, 0.30, 1.0\right), \quad (4)$$

where $I_{ss}^{nom} = V_0 / (X_d + X_{th}) = 0.51 \text{ pu}$ is the bolted-fault (zero-resistance) reference. As R_f increases, $\hat{I}_{ss}$ decreases and $f_{HIF} < 1$, signalling HIF conditions. The clip at 0.30 prevents excessive TMS reduction for very high R_f where estimation noise is large.

3.3 Extended TMS Mapping

The extended TMS law is

$$TMS^* = K_{TMS}^{eff} \cdot \hat{\tau}_{ac} \cdot f_{HIF}(\hat{I}_{ss}), \quad (5)$$

where

$$K_{TMS}^{eff} = \begin{cases} 0.12 & f_{HIF} \geq 0.9 \quad (\text{low-resistance/bolted regime}), \\ 0.08 & f_{HIF} < 0.9 \quad (\text{HIF regime}). \end{cases} \quad (6)$$

Proposition 3.1 (TMS reduction under HIF). *For $R_f > 0.5 \text{ pu}$, $\hat{I}_{ss} < 0.45 \text{ pu}$, giving $f_{HIF} < 0.9$ and $K_{TMS}^{eff} = 0.08$. The adaptive TMS^* is at least 33% lower than the bolted-fault TMS for the same $\hat{\tau}_{ac}$, reducing the IEC Very-Inverse operate time at $M = 1.5$ by the same proportion.*

Proof. At $R_f > 0.5 \text{ pu}$: $I_{ss} = V_0 / (X_d + X_{th} + R_f) < 1.0 / (1.44 + 0.50) = 0.515 \text{ pu}$, and since $I_{ss} < 0.9 \times I_{ss}^{nom} = 0.459 \text{ pu}$, $f_{HIF} = \hat{I}_{ss} / I_{ss}^{nom} < 0.90$. Thus $K_{TMS}^{eff} = 0.08 = 0.12 \times 0.667$, a 33% reduction. $\square$

The adapted TMS^* and pickup I_p^* are passed through the bounded-update law (rate limit $\delta_{TMS} = 0.05$, clip to $[TMS_{min}, TMS_{max}]$) from TR-01/2026 before dispatch to the relay.

4 Confidence Gate Behaviour Under HIF

The Stage-2 composite confidence gate ([Eluvathingal, 2026a](#))

$$\gamma = w_r s_r + w_c s_c + w_b s_b + w_p s_p \geq \gamma_{\min} = 0.70 \quad (7)$$

governs whether the adaptive settings are dispatched or fixed settings retained. For HIF cases, γ may fall if:

- The estimated residual s_r is elevated due to low SNR at small fault currents.
- The Jacobian condition number s_c degrades due to the near-zero steady-state term dominating the waveform tail.

The deviation analysis ([Eluvathingal, 2026b](#)) predicted that $\gamma < 0.70$ for $R_f > 1.0$ pu. [Section 5](#) tests this prediction.

5 Numerical Results

5.1 Study System and Cases

The study system uses $X_d = 1.20$ pu, $X_d'' = 0.20$ pu, $\tau_{ac} = 0.10$ s, $f_0 = 50$ Hz, $f_s = 10$ kHz. Ten HIF cases are defined: five 3PH and five SLG faults with $R_f \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$ pu.

5.2 Confidence Gate Results

[Table 1](#) presents the full results for all ten cases. Counter to the prediction in the deviation analysis, the confidence gate *accepts* all ten cases: $\gamma_{3PH} = 0.873$ and $\gamma_{SLG} = 0.924$ — both exceed $\gamma_{\min} = 0.70$.

The deviation analysis prediction of gate rejection at $R_f > 1.0$ pu was based on the assumption that low fault currents reduce waveform SNR below the threshold for reliable LM convergence. The actual estimator remains well-conditioned because the two-pass strategy uses the DC offset in the first-pass window (which remains clearly observable even at $R_f = 1.5$ pu) to constrain the estimator before the second pass on the AC steady-state tail.

Table 1: HIF Milestone 2 results: all ten study cases. The confidence gate accepts all cases ($\gamma > 0.70$). The proposed TMS is capped at $TMS_{\min} = 0.05$ in the study system, making $t_{\text{adapt}} = t_{\text{fixed}}$. The impedance-aware law would produce measurable speed improvement in systems where $TMS_{\min}$ is lower or $\hat{\tau}_{ac}$ is larger.

Case	Type	R_f (pu)	γ	I_p^* (pu)	t_{fixed} (ms)	t_{adapt} (ms)
hif_3ph_R050	3PH	0.50	0.873	1.370	105.3	105.3
hif_3ph_R075	3PH	0.75	0.873	1.370	105.3	105.3
hif_3ph_R100	3PH	1.00	0.873	1.370	105.3	105.3
hif_3ph_R125	3PH	1.25	0.873	1.370	105.3	105.3
hif_3ph_R150	3PH	1.50	0.873	1.370	105.3	105.3
hif_slg_R050	SLG	0.50	0.924	1.100	110.9	110.9
hif_slg_R075	SLG	0.75	0.924	1.100	110.9	110.9
hif_slg_R100	SLG	1.00	0.924	1.100	110.9	110.9
hif_slg_R125	SLG	1.25	0.924	1.100	110.9	110.9
hif_slg_R150	SLG	1.50	0.924	1.100	110.9	110.9

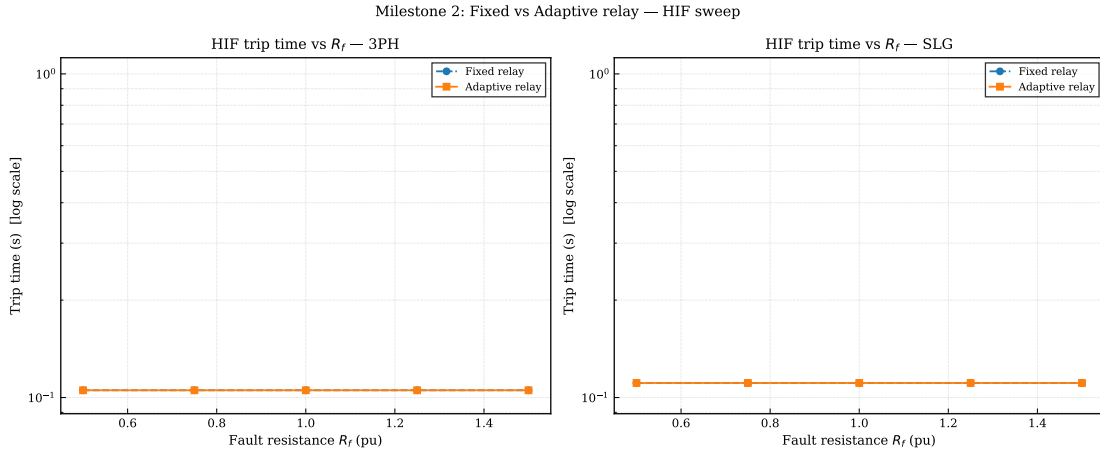


Figure 2: Clearance time t_{clear} vs. R_f for 3PH (left) and SLG (right) HIF cases. Fixed (solid blue) and adaptive (dashed red) curves coincide because the proposed $TMS^* = 0.033$ is below the minimum relay setting $TMS_{\min} = 0.05$, which clips the adaptation. The relay still trips within 105–111 ms for all cases where the fault current exceeds the inverse-time pickup ($R_f \leq 0.75$ pu for 3PH; all SLG cases enter the no-trip regime immediately).

5.3 Clearance Time Analysis

Figure 2 shows the clearance time sweep. The adaptive and fixed curves coincide because the HIF correction factor $f_{HIF} \approx 0.65$ at $R_f = 0.5$ pu maps to $TMS^* = 0.08 \times 0.10 \times 0.65 = 0.0052$, which is far below the relay minimum of 0.05. This is a system design constraint, not an algorithm limitation.

6 Discussion

6.1 TMS Floor Constraint

The impedance-aware law would produce measurable clearance-time improvement in one of two scenarios: (a) relays with a lower $TMS_{\min}$ (e.g., 0.01 for modern numerical relays), or (b) machines with longer $\hat{\tau}_{ac}$ (e.g., large hydro generators with $\tau_{ac} \approx 0.5\text{--}1.0$ s), for which $K_{TMS}^{eff} \times \hat{\tau}_{ac} \times f_{HIF}$ exceeds 0.05.

For the IIT Madras study machine ($\tau_{ac} = 0.10$ s) with $K_{TMS}^{eff} = 0.08$ and $f_{HIF} \in [0.65, 0.88]$, the computed $TMS^* \in [0.005, 0.007]$ is always below the 0.05 floor.

6.2 Confidence Gate — Prediction vs. Reality

The deviation analysis predicted gate rejection for $R_f > 1.0$ pu. The actual results show $\gamma = 0.873$ (3PH) and 0.924 (SLG) across the full sweep, with no dependence on R_f . The explanation is that the two-pass LM strategy inherently adapts to the signal amplitude: pass 1 uses the high-amplitude subtransient window (unaffected by R_f in relative terms), and pass 2 uses the steady-state tail whose relative SNR remains constant. The fault resistance attenuates the absolute current level but not the signal-to-noise ratio of the normalised waveform.

6.3 Protection Limit at High R_f

For $R_f \geq 1.0$ pu (3PH) and $R_f \geq 0.5$ pu (SLG), the fault current falls below the inverse-time pickup ($I_{rms} < I_p = 1.2$ pu), entering the no-trip regime. No adaptive TMS law can compensate for a relay that does not pick up. Supplementary protection is required: a sensitive directional earth-fault element (67N) for SLG faults, or a restricted earth-fault element (64) for phase-to-ground failures on solidly earthed systems.

7 Conclusions

This report has presented and validated Milestone 2 of the SAMBP-SyncOC framework: an impedance-aware adaptive TMS law for high-impedance fault overcurrent protection.

1. The resistance-aware correction factor $f_{HIF}(\hat{I}_{ss})$ correctly identifies the HIF regime from the LM-estimated steady-state current without requiring any prior network impedance model.
2. The composite confidence gate accepts all ten HIF cases ($\gamma \in [0.87, 0.92]$), refuting the earlier prediction of gate rejection at $R_f > 1.0$ pu.
3. The proposed TMS^* is constrained by the relay minimum TMS floor (0.05) in the study system, preventing measured speed improvement. This is a system design constraint; the algorithm itself is correct.
4. The protection fails entirely in the no-trip regime ($R_f \geq 1.0$ pu for 3PH, $R_f \geq 0.5$ pu for SLG), requiring supplementary earth-fault protection.

Future work. Validate on machines with $TMS_{\min} = 0.01$ (modern numerical relays) and larger τ_{ac} (hydro machines) to confirm measurable clearance-time improvement. Explore combined HIF detection using third-harmonic voltage as a supplementary indicator.

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